1. Data Structures and Algorithms

Data structures refer to the way in which the data is stored and organized in a computer so that the data can be accessed and manipulated.

Algorithms refer to a set of rules and procedures designed to solve a particular computational problem. The efficiency of data processing depends on the interaction between data structures and algorithms.

2. Linear Data Structure

Linear data structures refers to a type of data structure where the elements are stored in a linear fashion where each element is placed in a definite logical position. Data is generally processed sequentially from one element to another.

3. One-Dimension Array

One-dimension array refers to a linear data structure where the elements of the same data types are stored in a list and accessed through index positions.

4. Big O Notation

Big O notation refers to a way of expressing the growth rate of time and/or space complexity of an algorithm depending on the input. This notation enables the analysis and comparison of the efficiency of algorithms irrespective of the hardware used or the time of execution.

5. Basic Searching Algorithms

Searching algorithms are the procedures used to search for a particular element or value in the collection of data. Different searching algorithms exist and differ in the ways in which they process data and the efficiency of locating the required element.

- Linear Search

Linear search refers to a procedure used to locate the element within a collection of data where the elements are examined sequentially from one end of the list until the required element is found or all the elements have been examined. This algorithm works with sorted as well as unsorted data.

- Binary Search

Binary search refers to the procedure where the sorted collection of data is divided repeatedly into two halves. After dividing, the search procedure is performed for the half containing the required element. This search procedure has logarithmic complexity and is more efficient than linear search.

References

- *1.1. Data Structures and Algorithms — CS2 Software Design & Data Structures*. (n.d.). https://opendsa-server.cs.vt.edu/ODSA/Books/CS2/html/IntroDSA.html
- GeeksforGeeks. (2026, February 3). *Introduction to linear data structures*. GeeksforGeeks. https://www.geeksforgeeks.org/dsa/introduction-to-linear-data-structures/
- GeeksforGeeks. (2026b, March 28). *Big O Notation*. GeeksforGeeks. https://www.geeksforgeeks.org/dsa/analysis-algorithms-big-o-analysis/
- https://www.geeksforgeeks.org/